

Script -Controlled-ACL Restrict Record

Access Based on Field value

❖ Introduction

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❖ Project Overview

***This project focuses on using a script controlled Access Control List (ACL) in ServiceNow to restrict record access based on a specified field value. The ACL script checks the field value and controls whether the user is allowed to access the record. The main purpose of the project is to provide controlled and secure access to ServiceNow records**

❖ Milestone-1 Creation of Users and Roles

a)User Creation:

- 1. Log in to the ServiceNow instance with admin access.**

2. Create a test user:

- ☐ **Navigate to User Administration → User**
- ☐ **Click New**
- ☐ **User ID: EEE User**
- ☐ **First name: EEE**
- ☐ **Last name: User**
- ☐ **Email: eeeuser@gmail.com**
- ☐ **Save the record**

ServiceNow Developers x Users | ServiceNow x + Ask Gemini

dev302430.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253e%253dEEE%26sysparm_...

serviceNow All Favorites History Workspaces Users ☆ Search

Users Name Search Actions on selected rows... New

All > Name >= EEE

User ID	Name	Email	Active	Created	Updated
EEE User	EEE User	eeeuser@gmail.com	true	2026-09-23 05:36:47	2026-09-23 05:36:47
efren.baucher	Efren Baucher	efren.baucher@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:59
elaine.renzi	Elaine Renzi	elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-09-12 10:30:57
eldon.sutch	Eldon Sutch	eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-09-12 10:31:05
eli.bettner	Eli Bettner	eli.bettner@example.com	true	2012-02-17 19:04:51	2026-09-12 10:31:00
elisa.gracely	Elisa Gracely	elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-09-12 10:30:57
eliseo.wice	Eliseo Wice	eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:58
ella.pahnke	Ella Pahnke	ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-09-12 10:31:03
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-09-12 10:31:04
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-09-12 10:30:56
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-09-12 10:31:02
emerv.reek	Emerv Reek	emerv.reek@example.com	true	2012-02-17 19:04:49	2026-09-12 10:30:58

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b) Roles Creation

- **Navigate to User Administration → Roles**
- **Click New**
- **Name: Enter role name as bb1**

- **Description: Brief role purpose**
- **Click Submit**
- **Create other roles also like bb2,bb3,bb4.**
- **Assign all the roles to the EEE User**

The screenshot shows the ServiceNow interface for the 'User - EEE User'. The 'Roles' tab is selected, displaying a table of roles assigned to the user. The table has columns for Role, State, Inherited, and Inheritance Count.

Role	State	Inherited	Inheritance Count
bb4	Active	false	
bb1	Active	false	
bb2	Active	false	
bb3	Active	false	
snc_required_script_writer_permission	Active	true	1

❖ Milestone 2 :Tables Creation

1. In the Application Navigator, search for system Definition → Tables
2. Click New and create a custom table with the following Details.
 - **Label: Institution Details**
 - **Name: u_institution_details**
 - **Extends for : false**
3. Click Submit

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Application ⓘ

* Name Remote Table

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40	40	false
Email	String	(empty)	40	40	false
Sys ID	Sys ID (GUID)	(empty)	32	32	false
Created by	String	(empty)	40	40	false

4. Add the following fields to the u_institution_details table:

- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String

5. Save the Table

Table - Institution Details

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updated	Date/Time	(empty)	40		false
✗ Email	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
✗ Faculty Name	Reference	User	32		false
✗ Student Name	Reference	User	32		false
Created	Date/Time	(empty)	40		false
✗ Description	String	(empty)	40		false
✗ Student Roll Number	String	(empty)	40		false
Updated by	String	(empty)	40		false
✗ Branch	Choice	(empty)	40		false
Updates	Integer	(empty)	40		false
✗ Phone Number	String	(empty)	40		false
+ Insert a new row...					

Delete Update Delete All Records

❖ Milestone 3 : Creation of Access Control Lists(Read ACL)

1. In the Application Navigator, search for System Security →

Access control (ACL)

2. Elevate “Security_admin” role

3. Click New to create a new ACL.

4. Enter the following details:

- ☐ **Type: record**
- ☐ **Operation: read**
- ☐ **Name: u_institution_details**
- ☐ **Active: true**
- ☐ **Advanced: true**

5. In the Requires role related list, add:

- ☐ **bb1 (custom role)**

6. Add Data condition as Branch is EEE

7. In the Script field, add the following script:

```
(function () {  
// Allow admin users full access  
if (gs.hasRole('admin')) {  
    return true;  
}  
// Allow only EEE branch users to see EEE records  
if (gs.hasRole('bb1'))  
{  
    return true;  
}  
// Deny access for all others  
return false;  
})();
```

8. Click Submit to save the ACL.

9. Impersonate EEE users and open the Student Records list.

10. Copy the table name u_institution_details.list/LIST and paste in the all menu

11. Impersonate Admin and open the Student Records list.

Verify :

- **User with bb1 role can view only EEE branch records**
- **User without the role cannot view any records**
- **Admin user can view all records regardless of branch**

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The top navigation bar includes 'serviceNow', 'All', 'Favorites', 'History', and a search bar. The main header is 'Access Control - u_institution_details'. The configuration form includes the following fields:

- Type: record
- Operation: read
- Application: Global
- Active: ☒
- Admin overrides: ☐
- Advanced: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- Name: u_institution_details
- Description: Allow read for records in u_institution_details, for users with role bb1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE^EQ) evaluates to true.

Below the form is a 'Conditions' section with a blue box containing the following text:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

There is a link for [More Info](#).

Below the conditions is a 'Requires role' section with a table:

Role
bb1

At the bottom, there is a 'Security Attribute Condition' section.

❖ Milestone 4 : Creation of Access Control Lists(Create ACL)

1. Click New to create a new ACL(Create)

2. Enter the following details:

- ☐ **Type: record**
- ☐ **Operation: create**
- ☐ **Name: u_institution_details**
- ☐ **Active: true**

3. In the Requires role related list, add:

- ☐ **bb2 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify:

- **User with bb1 &bb2 roles can view only EEE branch records b**

New Button

The screenshot shows the ServiceNow interface for configuring an Access Control Rule. The rule is named 'u_institution_details' and is set to 'create' operation on 'record' type. The application is 'Global', and it is active. The decision type is 'Allow If'. The protection policy is '-- None --'. The description is 'Allow create for records in u_institution_details, for users with role bb2.'.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
bb2

❖ Milestone 5 : Creation of Access Control Lists(Write ACL)

1. Click New to create a new ACL(Write)

2. Enter the following details:

- **Type: record**
- **Operation: Write**
- **Name: u_institution_details**
- **Active: true**

3. In the Requires role related list, add:

- **bb3 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify : ● User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button

The screenshot shows the ServiceNow interface for configuring an Access Control List (ACL) for the resource 'u_institution_details'. The configuration is as follows:

- Type:** record
- Operation:** write
- Application:** Global
- Active:** ☒
- Admin overrides:** ☒
- Advanced:** ☐
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** u_institution_details
- Description:** Allow write for records in u_institution_details, for users with role bb3.

Conditions:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
bb3

Security Attribute Condition

❖ Milestone 6 : Creation of Access Control Lists(Delete ACL)

1. Click New to create a new ACL(Delete)

2. Enter the following details:

- ☐ **Type: record**
- ☐ **Operation: Delete**
- ☐ **Name: u_institution_details**
- ☐ **Active: true**

3. In the Requires role related list, add:

- ☐ **bb4 (custom role)**

4. No need to add any Data condition

5. Save the ACL Record

Verify :

● **Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.**

servicenow All Favorites History : Access Control - u_institution_details Search

Access Control u_institution_details

Type record Application Global

Operation delete Active ☒

Admin overrides ☒ Advanced ☐

Decision Type Allow If

Protection policy -- None --

Name u_institution_details

Description Allow delete for records in u_institution_details, for users with role bb4.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role 1 to 1 of 1

Role
bb4

Security Attribute Condition

❖ Milestone 7 : Conclusion

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

ServiceNow Developers

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dev302430.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

YouTubeMapsGmailPrompt Design in A...Learning Course - S...

servicenow

AllFavoritesHistory

Institution Details s

Branch

Search

Actions on selected rows...

New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

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Search

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